

AY25 PROGRAM ASSESSMENT REPORT

v2026.1

Tracking Data

CIM code (e.g., ME-CM)	MS-ARCE
Program Name	Master of Science in Architectural Engineering
Assessment Year	2024-2025
Program Assessment Coordinator	Brent Stephens
Year of Current Assessment Plan	2024-2025
No Assessment This Year per Plan	☐
LO(s) Assessed	2
Presented at program authority meeting date: <i>Please attach minutes</i>	n/a

Methodology

The Master of Science in Architectural Engineering is a research- and thesis-based graduate degree program oriented toward students who wish to develop more knowledge about, and conduct research in, the design, construction, and operation of buildings and their systems (e.g., mechanical, electrical, plumbing, lighting, and more). The degree program is intended for students who want to perform deeper investigations in building science, building performance, energy efficiency, and sustainable buildings.

Learning objectives for this program are for students to:

1. Demonstrate advanced knowledge in and an ability to apply technical skills to fundamental problems in architectural engineering
2. Formulate research questions, conduct literature reviews, and utilize relevant tools to advance knowledge in architectural engineering

Learning Objective #1: Students will be able to demonstrate advanced knowledge in and an ability to apply technical skills to fundamental problems in architectural engineering.

- Artifact used to evaluate student learning:
 - Performance on assignments and exams in CAE 513 Building Science (using performance indicator ranging from 0 to 3)
 - Performance indicators are summarized topically into 4 key subject areas in architectural engineering: a) building physics, b) psychrometrics, c) ventilation and IAQ, and d) load calculations and energy balances
- Source for evaluation artifact (e.g., course(s), assignment(s), or other sources):
 - CAE 513 Building Science assignment and exam grades
 - Assignments and exams that cover:
 - 1a: Building physics, heat transfer, and energy concepts (HW 1, HW 2, Exam 1)

- 1b: Psychrometric principles and applications (HW 3, HW 4, Exam 2)
 - 1c: Ventilation and indoor air quality (HW 5)
 - 1d: Heating/cooling load calculations and energy balances (HW 6, Final Exam)
- Rationale for artifact and source:
 - The course is a required course in the program; it covers a breadth of fundamental subjects in architectural engineering; each of the problem sets used for evaluation allow for systematic evaluation of performance indicators related to this LO
- Semester(s) in which artifacts are collected:
 - End of each academic year
- Sample size:
 - Each student in the program who took CAE 513 Building Science
- Data to report:
 - Results are reported as statistical distributions (i.e., histograms) and summary statistics for the sample of students in each year on each performance indicator
- Month/year of assessment:
 - End of each academic year
- Names/titles of evaluators:
 - Academic Program Coordinator, Program Director, and Department Chair

Learning Objective #2: Students will be able to formulate research questions, conduct literature reviews, and utilize relevant tools to advance knowledge in architectural engineering.

- Artifact used to evaluate student learning:
 - Performance on final written report for CAE 513 Building Science (using performance indicator ranging from 0 to 3)
- Source for evaluation artifact (e.g., course(s), assignment(s), or other sources):
 - CAE 513 Building Science final written report instructor evaluation
- Rationale for artifact and source:
 - The course is a required course in the program; written report allows for exploration of depth of knowledge and ability to use relevant tools within a single subject in architectural engineering
- Evaluation rubric:
 - Clear justification and explanation of your ideas and motivation (20%)
 - Comprehensive literature review and quality/quantify of references (20%)
 - Follows specified format (10%)
 - Technical foundation for your project (20%)
 - Depth of investigation (30%)
- Semester(s) in which artifacts are collected:
 - End of each academic year
- Sample size:
 - Each student in the program who took CAE 513 Building Science
- Data to report:
 - Results are reported as statistical distributions (i.e., histograms) and summary statistics for the sample of students in each year on each performance indicator
- Month/year of assessment:
 - End of each academic year
- Names/titles of evaluators:
 - Academic Program Coordinator, Program Director, and Department Chair

Results

Data table

Only 1 students who took CAE 513 was enrolled in the MS-ARCE program in Fall 2024 (30 graduate students in total took CAE 513 from a variety of other majors). Table 1 shows results from the 2 students in the program across LOs #1 and #2, as well as distributions of performance indicators for LO #1 subdivided topically into four key subject areas of (a) building physics, (b) psychrometrics, (c) ventilation and IAQ, and (d) load calculations and energy balances). Scores range from 0 to 3. Definitions of proficiency levels are approximately as follows: >2.85 (proficiency), 2.5-2.85 (meets expectations), <2.5 (does not meet expectations).

Table 1. Results from evaluation rubric applied to learning objectives (LOs) #1 and #2 using data from CAE 513 Building Science. Scores range from 0 to 3.

Student	LO #1					LO #2 (Report)
	1a (HW1&2, Exam1)	1b (HW3&4, Exam2)	1c (HW5)	1d (HW6, Exam3)	LO #1 Overall	
1	3.00	2.93	3.00	3.00	2.98	2.70

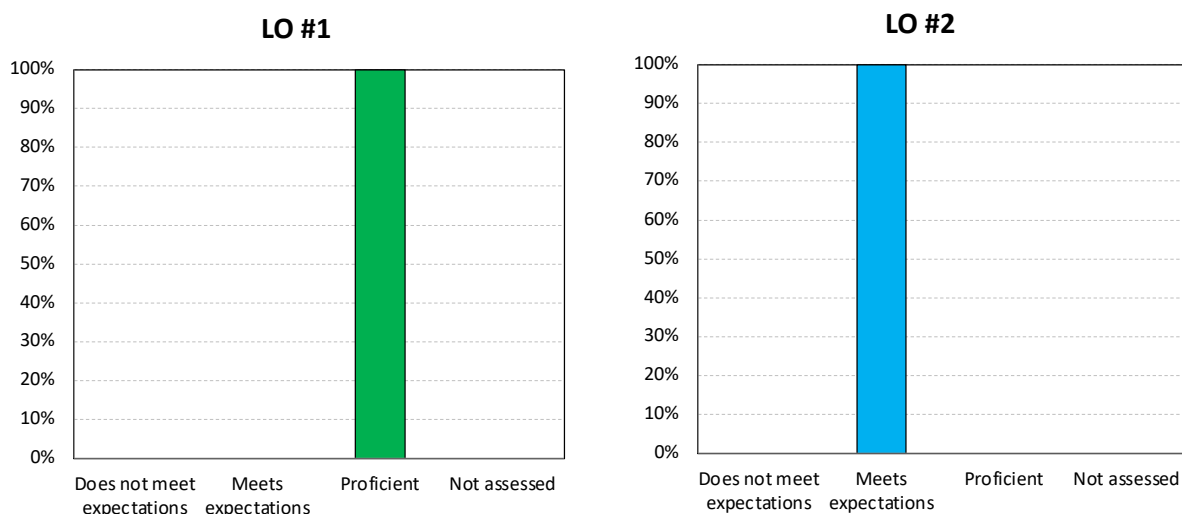


Figure 1. Distribution of proficiency levels for LO #1 and LO #2 for this academic year.

Description of findings

Key findings from this academic year and cohort include:

- The one student that was sampled met expectations for LO #1 and was proficient with an overall score of 2.98 out of 3.0, demonstrating advanced knowledge in and an ability to apply technical skills to fundamental problems in architectural engineering, assessed via homework problems

and exam problems that required students to apply knowledge to solve problems. The student took the in-person section.

- Among the four key subject areas for assessing LO #1, no subject areas had low scores below 2.5 and thus there were no subjects of concern in this student.
- Performance on LO #2 met expectations (2.70 out of 3.0), demonstrating an ability to formulate research questions, conduct literature reviews, and utilize relevant tools to advance knowledge in architectural engineering through the final project report, which includes components of literature review, problem setup, analysis, and discussion.

Prior CI plan impact

In the previous assessment report, the instructor (Stephens) decided to minimally adjust the course based on findings and program advisors committed to keeping a close eye on students, especially for those coming from non-engineering backgrounds. In fact, advisors attempted to improve preparation by students from non-engineering backgrounds by incorporating stricter prerequisite requirements at admission. We continue to execute these continuous improvement actions.

This cycle's CI plan

1. After making several improvements in the last cycles, the instructor (Stephens) has again decided to minimally adjust the course based on these findings.
2. Program advisors will continue to keep a close eye on students, especially for those coming from non-engineering backgrounds (per previous years' findings).
3. Continuing with last year's plan, advisors will attempt to improve preparation by students from non-engineering backgrounds by incorporating stricter prerequisite requirements at admission. We are paying closer attention to transcripts in the application review process, including scrutinizing what "engineering" means at different universities across the globe (i.e., in many countries architectural engineering lacks sufficient mathematics or physics backgrounds). In these cases, we now accept with deficiency course requirements of CAE 208 Thermodynamics and/or CAE 302 Fluid Mechanics, plus any required engineering mathematics courses that are missing.
 - a. None of the students in this cohort needed these prerequisite deficiency courses so this point is irrelevant for these students.
4. In discussions with core program faculty (Stephens, Heidarinejad, and Corradi), a need arose to reevaluate our definition of learning objectives for the program and also which course(s) we use for assessment for 2025-2026 and beyond.
 - a. One issue that arose is that our learning objectives do not currently articulate an important focus on the implementation of principles of integrated sustainable design for high-performance buildings. We will explore ways to introduce this LO in the future.
 - b. Another issue that arose is that since this is a research- and thesis-based degree program, we could utilize a thesis defense rubric to support program assessment. The CAEE department developed a rubric for use and will begin deploying in the next academic year.
 - c. Another issue that arose is that by using deliverables in CAE 513 as our primary means of assessment, we are missing some students. Some students in recent years have come in with backgrounds in topics similar to those in CAE 513 and might not take it, and, more importantly, many of the students in the program are co-terminal/accelerated

master's students pursuing both a bachelor's and master's simultaneously in a 4+1 program. Those students take an undergraduate version of CAE 513 (CAE 331) and don't end up taking CAE 513 given significant overlap. For future years, we will consider moving to other required course(s) for assessment (i.e., CAE 526, CAE 556, and/or CAE 557) given their larger sample sizes (they cannot be substituted).

- d. As such, LOs for future years may evolve to a form of the following:
 - i. Demonstrate advanced knowledge in and an ability to apply technical skills to fundamental problems in architectural engineering
 - ii. Demonstrate an ability to implement principles of integrated sustainable design for high-performance buildings
 - iii. Formulate research questions, conduct literature reviews, and utilize relevant tools to advance knowledge in architectural engineering